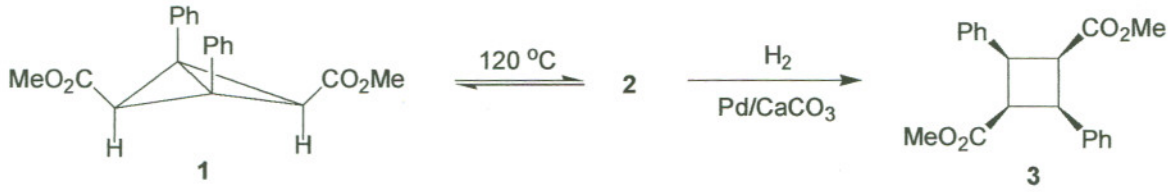
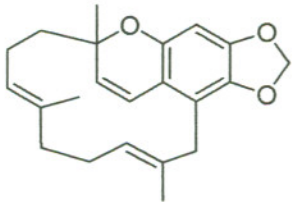
Organic Cume
12/5/03

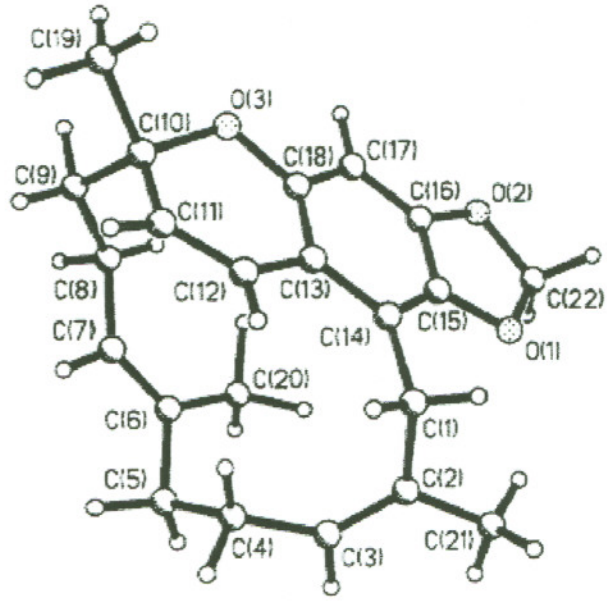
1) It has been proposed that the pure, crystalline bicyclobutane **1** undergoes thermal equilibration at 120 °C to yield roughly a 1:2 mixture of **1** and its isomer **2**. The new, major isomer can be isolated by chromatography, and is likewise equilibrated to the same mixture.

In any of several solvents, pure **2** shows a sharp singlet (2H) at about δ 4.6, and a sharp singlet (6H) at about δ 3.6. Moreover, mild catalytic reduction of **2** (10 %Pd-$CaCO_3$/H_2) gave exclusively the all-*cis* cyclobutane diester **3**. Assuming that the structure of **1** is correct (which it is), propose a unique structure for **2** that fits all the above data. (JACS 1969, 4612).



2) The natural product smenochromene A, isolated from a sponge off the Seychelle Islands, has the structure shown below and to the right.





a) Assign the following ^{1}H NMR signals:

δ 6.38 (d, 1H, J = 10 Hz)
δ 6.25 (s, 1H)
δ 5.35 (d, 1H, J = 10 Hz)

b) There are three singlets with integral = 3 located at δ 1.88, 1.16, and 1.42. It was shown that the signal at 1.42 was due to the methyl at C19. Assign the other two and briefly explain your choice.

c) How many triplets would appear in the proton-coupled ^{13}C spectrum?